Investigating the Impact of Generative AI Tutoring Styles on Student Knowledge Acquisition and Confidence Calibration

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1 Introduction

The rise of Generative AI (GenAI) and its capabilities has recently become a major topic of discussion in the education industry [**?**]. Ever since the release of ChatGPT by OpenAI in November 2022, the education landscape has evolved tremendously [**?**]. Whereas institutions initially banned the use of GenAI, the trend has shifted to embracing and integrating GenAI into pedagogy [**?**]. This transition is supported by GenAI's ability to deliver personalized learning and customized feedback based on individual student needs [**?**]. By utilizing GenAI, students are now able to focus on higher-order skills such as critical thinking and creativity, while AI scaffolds and automates repetitive tasks [**?**, **?**].

However, if used unconsciously, GenAI can present a significant set of challenges. As stated before, GenAI is able to take over repetitive tasks [**?**], but this could also lead to "cognitive offloading," where students outsource critical cognitive tasks to the AI [**?**]. It becomes even more concerning as AI could also create a "productivity paradox," where students experience increased productivity but lack depth in their critical analysis [**?**]. For example, due to convenicence and speed that AI summaries provide, students tend to prioritize using AI summaries over verifying the underlying sources [**?**]. Despite the faster drafting process, the final result ended up being lower quality, with generic essays and a lack of critical analysis [**?**, **?**]. This raises the importance of AI literacy, as studies shows that students with higher AI literacy tend to take a collaborative approach by planning and evaluatiing AI outputs, which consequently leads to better outcomes compared to students with lower AI literacy which tends to over-rely on the AI [**?**]. Additionally, being over relaint on AI can reduce face-toface interaction, leading to a decline in interpersonal skills and emotional intelligence which consequently leads to social anxiety and isolation [**?**].

As mentioned above, over-reliance on AI is a significant concern that must be addressed. AI tend to generate text with a convincing tone, giving an illusion that the output is always true, causing users to overlook hallucinations [**?**, **?**]. Addressing this, students are required to possess foundational knowledge before adopting the tool to ensure they are able to critically validate the information provided as the human in the loop is responsible for the final output [**?**, **?**]. Thus, students must be taught to verify AI outputs, while educators should create critical thinking assessments that can't be replicated by AI, such as video presentations or pitches [**?**, **?**].

To combat the illusion of truth, AI tools must be designed with guardrails that encourage cognitive engagement rather than handing out direct answers [**?**]. In the context of learning, the design of the AI interaction itself is more important than the it's capabilities [**?**]. Evidence shows that the use of scaffolding techniques, such guiding hints and follow-up questions are able to significantly improve student's learning outcomes. AI chatbots with these guardrails are referred to as Socratic chatbots, designed to prevent giving out direct answers to encourage self-reflection and deep understanding of the topic [**?**, **?**, **?**]. When compared to default GenAI models, Socratic chatbots are able to produce 79% more learning improvments than default GenAI models (127% vs. 48%) in practice assesments [**?**]. Additionally, unike the default model where students experience learning loss when the AI is removed, students did not experience any learning loss when the Socratic chatbot is removed [**?**].

2 Literature Review

2.1 Generative AI in Education

2.2 Learning strategies: direct, Socratic, and learning-by-teaching

2.3 Confidence calibration and psychological impacts

2.4 Research gap

3 Research Question

Research question: How do direct-answer, Socratic, and reverse-tutoring (protégé effect) generative AI tutoring styles compare in terms of (a) student knowledge acquisition and (b) confidence calibration?

Hypothesis: Reverse tutoring (student teaches the AI as a naive tutee) will lead to better knowledge mastery and better alignment between confidence and actual competence than both the Socratic chatbot and the direct-answer chatbot, because teaching forces learners to organise knowledge and exposes gaps (protégé effect), reducing the illusion of competence while supporting deeper learning.

Optional sub-questions: Which style best supports long-term retention? Do self-efficacy and perceived difficulty differ across conditions?

4 Research Plan

4.1 Methodology

4.2 Data and experiment

4.3 Participants and ethics

4.4 Timeline

5 Conclusion

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